MGU-UGP (HONOURS) Regular Examination, October 2025

Third Semester

25900264 Discipline Specific Elective

MG3DSEBOTA05- BIOINSTRUMENTATION AND QUALITY ANALYSIS 1

PART A

Answer in one or two sentences. Answer **any 10** questions. Each question carry **2 marks**

1. What is the use of hot plate? [K] / [CO1]

To heat solutions or to dissolve compounds without use of direct flame

2. You are provided with 1 M Tris Buffer. How will you dilute this to make 500 ml 0.1 M buffer? [U] / [CO2]

N1V1 = N2V2 N1 = 1M; V1 = X N2 = 0.1 M, V2 = 500 ml

X = 50 ml. Take 50 ml stock solution in a 500 ml standard flask and make up this to 500 ml using distilled water.

3. What is the key difference between the equivalence point and the endpoint in a titration? [K] / [CO3]

Equivalence point: The point in a titration where the amount of added titrant is chemically equivalent to the amount of analyte in the sample, based on the reaction's stoichiometry. It is a theoretical, chemical concept. **End point:** The point in a titration where a visible change, most commonly a color change in the indicator, is observed. It is an experimental, practical observation.

4. What are the detectors used in Atomic absorption spectrophotometer? [K] / [CO4]

Photomultiplier tube (PMT). -It is a highly sensitive device that converts incident light into an electrical signal for measurement. Modern instruments increasingly use solid-state detectors like Charge-Coupled Devices (CCDs) and Charge-Injection Devices (CIDs)

5. List out the uses of conical flasks [K] / [CO1]

Titration, media preparation and autoclaving, culture flask, heating without much loss of solvents

6. What is the composition of carbonate-bicarbonate bufffer? [K] / [CO2]

Sodium carbonate and sodium bicarbonate. Sodium carbonate (weak base), sodium bicarbonate (weak acid).

7. Give any two applications of gravimetry in food analysis. [U] / [CO3]

Determination of fat content: Often done using a form of particulate or extraction gravimetry. Determination of moisture content: This is a common application of volatilization gravimetry

8. What is the use of a Colorimeter? [K] / [CO4]

to measure the concentration of a colored substance in a solution by measuring how much light is absorbed at a specific wavelength. Water quality testing, clinical and medical analysis, food and beverage industry, analyzing biochemical samples, and industrial quality control for products like paints and textiles.

9. Describe air displacement micropipette [K] / [CO1]

Adjustable micropipette that operate by piston-driven air displacement. Piston pushed down and then moved upwards creates a vacuum and the liquid around tip enter into the vacuum.

10. List out any two importance of pKa value [K] / [CO2]

pH at which concentration of conjugate base and un-dissociated acid remains the same. Calculation of pH if we know the concentration of conjugate base and un-dissociated acid. Determination of isoelectric point. To calculate pH of buffer and to determine the buffering capacity

11. Describe heat transfer mechanism in laboratory oven [K] / [CO1]

Convection- circulating hot air transfer heat to the item. Conduction- heat penetrates from surface to inwards. Radiation- by radiation from heating coil and heated oven wall

12. What is meant by strong acid? [K] / [CO2]
Completely dissociate to form ions. Have Ka value greater than 1 and pKa value less than 0

PART B

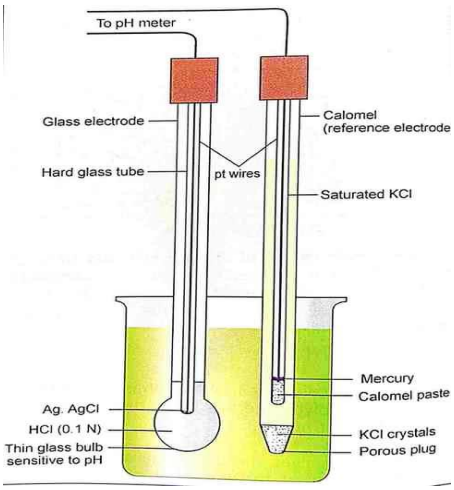
Short Answer Type Questions. Answer **any 8** questions. Each question carry **5 marks**

13. Recall the concepts of molarity and ppm [K] / [CO1]

Molarity- Number of moles of solute per litre of total solution. Molarity = Moles of solute/Volume of solution in litres. Moles of solute = mass of solute (g)/molar mass of solute.

PPM – Parts per Million. Amount of solute per one million parts of the total solution. Divide mass or volume of the solute by the total mass or volume of the solution and then multiply the result with one million (solute in mg, volume of solution in microliter)

14. Draw a labelled diagram of pH meter [U] / [CO2]



15. Write a brief note on safety protocols to be followed during acid-alkali titration. [K] / [CO3]

Wearing personal protective equipment (PPE) such as goggles, gloves, and a lab coat. Working in a well-ventilated area or fume hood. Handling chemicals carefully to avoid splashes. Knowing emergency procedures, like immediate rinsing with water for skin or eye contact. Dispose of chemicals in designated waste containers according to lab rules

16. Explain briefly electromagnetic spectrum. [U] / [CO4]

The full range of all electromagnetic radiation, ordered by frequency or wavelength. The spectrum is divided into separate bands, each with its own distinct name for the electromagnetic waves within it. From low to high frequency, these are: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays. The electromagnetic waves in each of these bands have different characteristics.

17. Describe the concept of distilled water and ultrapure water

Distilled water- water purified by distillation. Removes impurities and dissolved salts, TDS less than 1 ppm. Working- heating, condensation and collection. **Ultrapure water**- highest grade water. Contain only hydrogen and oxygen, TOC less than 1 microgram per ml. Pretreatment, primary purification (RO, degasification, UV treatment), ultrafiltration

18. What is a buffer. What are the factors that affect buffering capacity?

A solution that can resist pH change on addition of small amounts of acid/alkali. Mixture of weak acid and its conjugate base or a weak base and its conjugate acid.

Factors affecting buffering capacity- Concentration of buffer component (Higher concentration - greater buffering capacity), ratio of acid to base (highest when concentration of acid and base components are equal), pH (highest when pH is equal to pKa), temperature and ionic strength

19. Give a brief note on methyl orange and bromothymol blue as visual indicators in acid-alkali titrations. [U] / [CO3]

Methyl orange and bromothymol blue are pH indicators that change color in acid-base titrations, but they are suitable for different pH ranges and types of reactions. **Methyl Orange**: Red in acidic solutions and yellow in basic solutions. pH range- 3.1-4.4. Best for titrations with an equivalence point in the acidic range, such as a strong acid being titrated with a weak base. **Bromothymol Blue:** Yellow in acidic solutions and blue in alkaline solutions. pH range- 6.0-7.6. Best for titrations where the equivalence point is near neutral pH, such as a strong acid with a strong base.

20. What is the principle of UV-Visible spectrophotometer? [K] / [CO4]

Beer-Lambert Law – The absorbance of a solution is directly proportional to its concentration and the path length of the light through the solution. The instrument works by passing a beam of UV or visible light through a sample, and a detector measures the amount of light absorbed at each wavelength. This absorption causes electrons in the molecules to transition from a lower to a higher energy state.

21. What is an electronic balance? What are the precautions to be taken while using an electronic balance? [K] / [CO1]

An electronic balance is a high-precision digital scale used in laboratories to measure the mass of a substance with a high degree of accuracy. Unlike older mechanical balances, electronic balances use a force-measurement

transducer to convert the force exerted by the mass into an electrical signal, which is then displayed as a digital reading

Precautions- Ensure the balance is placed on a level, stable surface, away from vibrations, direct sunlight. Inspect the balance to ensure it is clean and free of any debris. Use the leveling screws to make sure the balance is level. Switch on the device and wait for 5 minutes to warm up the instrument. Select the correct unit of weighing. Never place the salt directly onto the balance pan, place the clean, dry weighing vessel. Tare to zero before weighing

22. Explain the measurement of pH using litmus paper and pH paper. [U] / [CO2]

Litmus paper- just whether solution is acidic or alkaline. Dip a piece of litmus paper in solution. Observe for colour change. Blue litmus turns red in acid and red litmus turns blue in alkali.

pH paper- measurement of ph. Dip a piece of litmus paper in solution or else take a clean glass plate, add a drop of solution. Cut a piece of litmus paper and place near the drop and using a clean rod push paper in to the rod. Observe for colour change.

PART C

Essay Type Questions. Answer **any 1** questions. Each question carry **10 marks**

23. Explain the various steps in gravimetric analysis. [U] / [CO3]

- **Sample preparation**: Weigh out a precise amount of the sample and dissolve it in a suitable solvent.
- **Precipitation:** Add a precipitating agent to the sample solution; this agent reacts with the analyte to form an insoluble precipitate
- **Digestion:** Allow the precipitate to digest, this allows for larger, purer crystals to form, which are easier to filter
- **Filtration:** Separate the solid precipitate from the liquid using a filter.
- **Washing:** Wash the precipitate on the filter paper to remove impurities
- **Drying or Ignition**: Dry the precipitate in an oven; this removes any remaining solvent and ensures the final product is pure
- **Weighing and Calculation**

24. Explain the principle of scattering spectroscopy with special reference to working and use of Raman spectrophotometer. [U] / [CO4]

When light interacts with a sample, some of it is scattered. While most light undergoes elastic Rayleigh scattering, a small fraction undergoes inelastic Raman scattering, where the scattered photons have a different energy (and therefore frequency) than the incident photons. This difference, known as Raman shift, corresponds to the vibrational modes of the molecule, providing a unique “fingerprint” for chemical identification in a Raman spectrophotometer.

- **Working**- Light source, sample interaction, spectrometer, filtering, dispersion, detection, and spectrum analysis
- **Uses**- Quality control, medical diagnostics, forensics, material identification, and in the pharmaceutical industry.